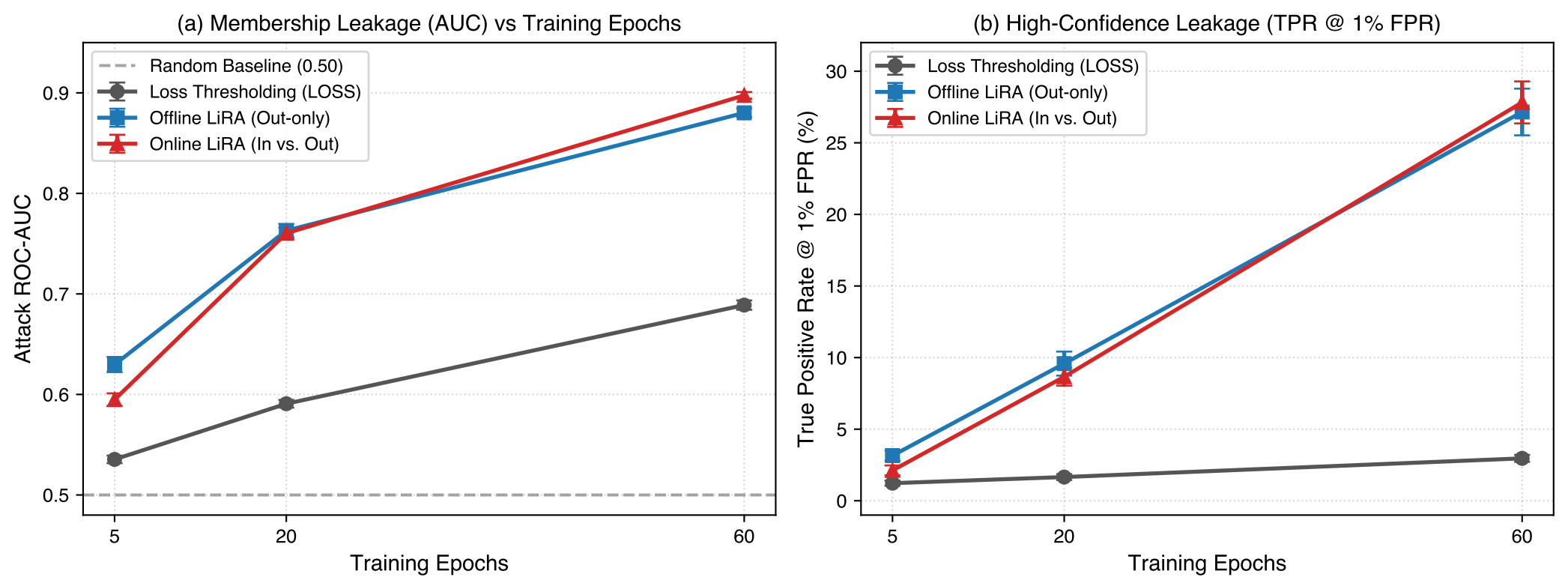
(a) Membership Leakage (AUC) vs Training Epochs
Random Baseline (0.50)
Loss Thresholding (LOSS)
Offline LiRA (Out-only)
Online LiRA (In vs. Out)
Attack ROC-AUC
0.9
0.8
0.7
0.6
0.5
5
20
60
Training Epochs
(b) High-Confidence Leakage (TPR @ 1% FPR)
Loss Thresholding (LOSS)
Offline LiRA (Out-only)
Online LiRA (In vs. Out)
True Positive Rate @ 1% FPR (%)
30
25
20
15
10
5
0
5
20
60
Training Epochs